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Intravascular catheter having an expandable incising portion and embolic protection device

Abstract

An intravascular device and methods for using the same are provided. The intravascular device includes an expandable portion with struts operable between a first position and a second position where the struts are moved outward from the first position. An incising element is provided at, and extends outward from, at least one of the struts. An embolic protection device covers at least a portion of the expandable portion.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 15/200,767 filed Jul. 1, 2016, the disclosures of which are incorporated herein by reference as if fully restated herein.

BACKGROUND OF THE INVENTION

(1) This invention relates in general to intravascular catheters, such as can be used during minimally invasive surgical procedures. In particular, this invention relates to an intravascular catheter having an expandable incising portion.

(2) Atherosclerosis is a chronic condition in which atheromatous plaque accumulates on the inner walls of a blood vessel. As a result, the blood vessel walls can become inflamed and, over time, may harden to form atherosclerotic lesions that cause a narrowing of the vessel lumen. In severe cases, the atherosclerotic lesions can rupture and induce the formation of thrombus (i.e., blood clots), which can prevent blood flow through the narrowed vessel lumen.

(3) There are known procedures and devices for treating or otherwise reducing the risks associated with atherosclerosis. For example, an angioplasty is a procedure in which a balloon catheter is inserted into a narrowed region of the vessel lumen via a delivery catheter. The balloon catheter includes a flexible tube having an inflatable balloon at an end thereof. Once positioned in the narrowed region, the balloon is inflated in order to dilate the narrowed vessel lumen. The pressure in the balloon is generally sufficient to compress the accumulated plaque. However, in some cases it would be desirable to fragment the atherosclerotic lesions. Thus, it would be desirable to provide an intravascular catheter having an expandable portion that can be selectively controlled by a user and adapted to create incisions in atherosclerotic material to facilitate fragmentation of the material during an angioplasty procedure.

(4) Embolism is a risk sometimes associated with many surgical procedures, such as angioplasty and the treatment of other peripheral artery diseases. A blood clot, air bubble, plaque fragment, or other embolism may be formed or be dislodged and travel through the patient's vascular system and cause damage. Embolic protection devices are sometimes placed in the patient's vascular system during surgical procedures in order to catch and remove emboli that may form or be dislodged. Use of such devices generally requires selection of the proper device, insertion and positioning of the device, performing the treatment, and removing said device. Proper design, inventory, and selection of embolic protection devices can be difficult as different treatment sites, procedures, and varying patient anatomy may require a healthcare provider to keep an inventory of many different devices to provide proper protection in the various conditions that may be encountered. Further, placement of these devices can be time consuming and expensive. Further still, the placed devices may be cumbersome and difficult to work around.

SUMMARY OF THE INVENTION

(5) This invention relates to an intravascular catheter device for use during a surgical procedure. The catheter device includes a catheter tube having an expandable portion with a plurality of struts each defining an outer surface. The expandable portion is operable between a closed position, wherein the expandable portion has a first diameter, and an opened position, wherein the expandable portion has a second diameter that is larger than the first diameter. An incising element is provided on the outer surface of at least one of the struts. The incising element has a sharpened edge that extends outwardly in a radial direction from the outer surface of the strut for creating an

incision in atherosclerotic material located within a blood vessel when the expandable portion is in the opened position.

(6) An embolic protection device may be integrated with the expandable portion. In exemplary embodiments, the embolic protection device is a mesh or film having a series of apertures located thereon. The embolic protection device may be positioned on the distal end of the expandable portion such that the embolic protection device forms a catch net when the expandable portion is placed in the opened position. The intravascular catheter device, and thus the net, may be positioned downstream of the treatment area with the net in the opened position such that the blood flow runs through the net once the intravascular device is positioned. The intravascular device may be retracted against the blood flow during treatment, thereby capturing any emboli that may be formed or dislodged as the intravascular device negotiates the treatment area. Once reaching the end of the treatment site, the surgeon may close the net, thereby trapping any captured emboli. The intravascular device, and thus the trapped emboli, may then be removed from the patient.

(7) Various aspects of this invention will become apparent to those skilled in the art from the following detailed description of the preferred embodiments, when read in light of the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a plan view of a catheter device that includes a handle assembly and a catheter tube having an expandable incising portion, in accordance with a first embodiment of this invention.
- (2) FIG. 2 is a cross-sectional side view of the handle assembly taken along section line 2-2 shown in FIG. 1 when the catheter device is in a first operating mode.
- (3) FIG. 3 is an enlarged cross-sectional side view of the catheter tube taken along section line 3-3 shown in FIG. 1 illustrating the expandable incising portion disposed within a blood vessel.
- (4) FIG. 4 is a cross-sectional end view of the expandable incising portion taken along section line 4-4 shown in FIG. 3.
- (5) FIG. 5 is a cross-sectional side view of the handle assembly taken along section line 2-2 shown in FIG. 1 when the catheter device is in a second operating mode.
- (6) FIG. 6 is an enlarged cross-sectional side view of the catheter tube taken along section line 3-3 shown in FIG. 1 illustrating the expandable incising portion in an opened position.
- (7) FIG. 7 is a cross-sectional end view of the expandable incising portion taken along section line 7-7 shown in FIG. 6.
- (8) FIG. 8 is an enlarged side view of a catheter tube having an expandable incising portion, in accordance with a second embodiment of this invention.
- (9) FIG. 9 is a side view of the catheter tube shown in FIG. 8 illustrating the expandable incising portion in an opened position.
- (10) FIG. 10 is a cross-sectional end view of the expandable incising portion taken along section line 10-10 shown in FIG. 9.
- (11) FIG. 11 is an enlarged side view of a catheter tube having an expandable incising portion, in accordance with a third embodiment of this invention.
- (12) FIG. 12 is a side view of the catheter tube shown in FIG. 11 illustrating the expandable incising portion in an opened position.
- (13) FIG. 13 is an end view of the catheter tube as shown in FIG. 12.
- (14) FIG. 14 is an enlarged side view of a catheter tube having an expandable incising portion, in accordance with a fourth embodiment of this invention.
- (15) FIG. 15 is a side view of the catheter tube shown in FIG. 14 illustrating the expandable incising portion in an opened position.

(16) FIG. **16** is an end view of the catheter tube as shown in FIG. **15**.

(17) FIG. **17** is a side view of the device shown in FIG. **8** with an exemplary embolic protection device installed thereon.

(18) FIG. **18** is a side view of the device shown in FIG. **17** with the expandable incising portion in the opened position.

(19) FIG. **19** is a side view of the device shown in FIG. **11** with the exemplary embolic protection device installed thereon.

(20) FIG. **20** is a side view of the device shown in FIG. **19** with the expandable incising portion in the opened position.

(21) FIG. **21** is a side view of the device shown in FIG. **11** with another exemplary embolic protection device installed thereon.

(22) FIG. **22** is a side view of the device shown in FIG. **21** with the expandable incising portion in the opened position.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(23) Referring now to the drawings, there is illustrated in FIG. **1** a catheter device, indicated generally at **10**, in accordance with this invention. The illustrated catheter device **10** is configured to treat or reduce the risks associated with atherosclerosis. In general, the catheter device **10** includes an expandable incising portion that can be inserted into a blood vessel and expanded to create incisions in atherosclerotic material that has accumulated on inner walls of the blood vessel. The incisions facilitate the fragmentation of the atherosclerotic material during a subsequent angioplasty or atherectomy procedure. Although the catheter device **10** will be described and illustrated in the context of treating atherosclerosis, it should be appreciated that the catheter device **10** can be used in any desired environment and for any desired purpose.

(24) Referring now to FIGS. **1** and **2**, the illustrated catheter device **10** includes a handle assembly, indicated generally at **20**. The illustrated handle assembly **20** includes an elongated, cylindrical handle body **21**. The handle body **21** may alternatively have any other shape that is suitable for easy handling by a surgeon. Further, the handle body **21** can be made from any suitably rigid material including, but not limited to, stainless steel or polymers.

(25) As shown in FIG. **2**, the illustrated handle body **21** defines an internal chamber **22**. A passage **23** extends into an end portion of the handle body **21** for communication with the internal chamber **22**. The handle body **21** further includes a slot **24** that extends through a side wall thereof for communication with the internal chamber **22**. The illustrated slot **24** may have any length or width as desired. As shown in FIG. **1**, an indicator **24A** may be provided on the handle body **21** adjacent to the slot **24**. For example, the indicator **24A** can be a visual scale or any other indicating means, the purpose of which will be explained below.

(26) The illustrated handle assembly **20** also includes a control member **25** that is supported on the handle body **21** for sliding movement within the slot **24**. For example, the control member **25** is movable between a forward position (shown in FIG. **2**), a rearward position (shown in FIG. **5**), or any position therebetween, which will be further explained below. As shown in FIG. **2**, the illustrated control member **25** includes a base portion **26** that is disposed within the internal chamber **22** of the handle body **21**. The base portion **26** may define an outer cross-sectional shape that generally corresponds with a cross-sectional shape of the internal chamber **22**, although such is not required. Alternatively, (or in addition), the control member **25** may be movably supported on the handle body **21** by a bearing, a bushing, a guide rail, or any other structural means. In other embodiments, the control member **25** may be supported for rotational movement, pivotal movement, or any other type of movement relative to the handle body **21**, the purpose of which will become apparent below. The visual indicator **24A**, described above, is configured to identify the relative position of the control member **25** with respect to the handle body **21**.

(27) The illustrated handle assembly **20** also includes a locking mechanism **27** that is configured to temporarily secure the control member **25** in a desired position, although such is not required. As

shown in FIG. 2, the illustrated locking mechanism 27 includes a plurality of protrusions that are spaced apart from one another along an inner surface of the slot 24. The control member 25 frictionally engages the protrusions to hold the control member 25 in the desired position. Alternatively, the locking mechanism 27 may be a threaded fastener, a pivotal latch, a push-button release, or any other mechanism that is configured to secure the control member 25 in a desired position.

(28) Referring now to FIGS. 1 through 3, the illustrated catheter device 10 also includes a catheter tube 30 that extends from the handle assembly 20. The catheter tube 30 is an elongated, flexible member having a proximal end that is secured to the handle assembly 20 and a distal end that extends therefrom. The catheter tube 30 can be made from any biocompatible material including, but not limited to, polyvinyl, polyethylene, nitinol, or stainless steel. Further, the catheter tube 30 can have any outer diameter, length, or wall thickness.

(29) As shown in FIG. 2, the proximal end of the catheter tube 30 is secured to the handle body 21 and communicates with the internal cavity 22 through the passage 23. The catheter tube 30 may be secured to the handle body 21 using a flanged connection, a fused connection, an adhesive, a press-fit connection, a threaded connection, or any other securing means. Alternatively, the catheter tube 30 may be secured to the handle body 21 using a connector or any other type of attachment device.

(30) As shown in FIGS. 1 and 3, an expandable portion 32 is provided on the distal end of the catheter tube 30. The illustrated expandable portion 32 is a cylindrical member having a longitudinal axis. The expandable portion 32 can be made from a generally resilient material that is able to flex between various positions, such as polyvinyl, polyethylene, nitinol, or stainless steel. The expandable portion 32 can be secured to the catheter tube 30 in any manner including, but not limited to, a fused connection, an adhesive, a press-fit connection, a threaded connection, or any other securing means. Alternatively, the expandable portion 32 can be integrally formed from the catheter tube 30. Further, the expandable portion 32 can have any outer diameter, length, or wall thickness.

(31) The illustrated expandable portion 32 has a pair of struts 34A and 34B. The illustrated struts 34A and 34B are separated by a pair of longitudinally extending slits 35A and 35B that extend through side walls of the expandable portion 32. As shown in FIG. 4, the slits 35A and 35B are equally spaced apart from one another around the circumference of the expandable portion 32 such that the struts 34A and 34B have the same circumferential widths, although such is not required. The struts 34A and 34B may have any length, circumferential width, or cross-sectional shape as desired.

(32) As shown in FIGS. 3 and 4, the illustrated expandable portion 32 also includes a pair of incising elements 36 that are respectively provided along outer surfaces of the struts 34A and 34B. The incising elements 36 can be atherotomes or other incising members having arcuate shaped sharpened edges, for example, that are configured to create incisions in atherosclerotic material as will be explained below. The illustrated incising elements 36 extend parallel with the longitudinal axis of the expandable portion 32 and outwardly in a radial direction therefrom. The incising elements 36 are equally spaced apart from one another around the circumference of the expandable portion 32. The expandable portion 32 may, however, have any number or configuration of incising elements 36 provided around the circumference thereof. Further, the incising elements 36 can have any cross-sectional shape, longitudinal length, or height and can be made from any suitable material including, but not limited to, tempered steel, stainless steel, high carbon steel, or ceramics. The incising elements 36 can be molded with the struts 34A and 34B or may otherwise be secured thereto in any manner such as, for example, using a welded or soldered connection, an adhesive, or any other fastening means.

(33) The distal end of the expandable portion 32 may optionally include a tip member 38. The illustrated tip member 38 has a generally conical shape that facilitates insertion of the catheter tube 30 within a blood vessel 50 (see FIGS. 3 and 4) and subsequent travel therethrough. The tip

member **38** may, however, have any desired shape. An aperture may axially extend through the tip member **38**, the purpose of which will be explained below. The tip member **38** can be integrally formed with the expandable portion **32** or may be secured thereto, such as with an adhesive or the like. Further, the tip member **38** can be made from any biocompatible material including, but not limited to, polyvinyl, polyethylene, nitinol, stainless steel, or polyether block amide.

(34) As shown in FIGS. **2** through **4**, the illustrated catheter device **10** also includes an inner sleeve **40**, although such is not required. The inner sleeve **40** is a flexible, tubular member that is supported for sliding movement within the catheter tube **30**, the purpose of which will be explained below. The inner sleeve **40** can be made from any biocompatible material including, but not limited to, polyvinyl, polyethylene, nitinol, stainless steel, or a woven material. Further, the inner sleeve **40** can have any outer diameter, length, or wall thickness. The inner sleeve **40** need not be a tubular member but may alternatively be a solid wire, a braided wire, or the like.

(35) As shown in FIG. **2**, a proximal end of the inner sleeve **40** extends from the catheter tube **30** and into the internal chamber **22** of the handle body **21**. The proximal end of the inner sleeve **40** is secured to the base portion **26** of the control member **25** for sliding movement therewith, the purpose of which will be explained below. The inner sleeve **40** can be secured to the base portion **26** by a flanged connection, a fused connection, an adhesive, a threaded connection, or any other securing means.

(36) As shown in FIG. **3**, the inner sleeve **40** extends through an entire length of the catheter tube **30**. A distal end of the inner sleeve **40** that is opposite the handle assembly **20** is secured to the tip member **38**, which is in turn secured to the expandable portion **32**. The inner sleeve **40** may be secured to the tip member **38** in any manner including, but not limited to, a fused connection, an adhesive, a fastener, or the like.

(37) Referring back to FIGS. **1** and **2**, the illustrated catheter device **10** also includes a protective sheath **42** that is supported for sliding movement along an outer surface of the catheter tube **30**, although such is not required. The protective sheath **42** can be made from any biocompatible material including, but not limited to, polyvinyl, polyethylene, nitinol, or stainless steel. Further, the protective sheath **42** can have any outer diameter, length, or wall thickness. The purpose of the protective sheath **42** will be explained below.

(38) The illustrated protective sheath **42** includes a flange **44** that facilitates sliding movement of the protective sheath **42** relative to the catheter tube **30**. The illustrated flange **44** is an annular member that is located at an end of the protective sheath **42** nearest the handle assembly **20**. The flange **44** can be integrally formed with the protective sheath **42** or may otherwise be secured thereto in any manner, such as with an adhesive or the like. It should be appreciated that the flange **44** can have any shape or may alternatively be configured in any manner to accomplish the functions described herein and below.

(39) The operation of the catheter device **10** will now be described with reference to FIGS. **1** through **7**. Referring initially to FIGS. **1** through **4**, the catheter device **10** is illustrated in a first operating mode. In the first operating mode, the control member **25** on the handle assembly **20** is located in the forward position relative to the handle body **21**. The inner sleeve **40** fully extends into the catheter tube **30** such that the expandable portion **32** is in a closed position, as shown in FIGS. **3** and **4**. In the closed position, the struts **34A** and **34B** are generally parallel with one another and with the inner sleeve **40**. The slits **35A** and **35B** (illustrated by the dashed lines in FIG. **3**) remain in a generally closed configuration. As such, the expandable portion **32** defines an initial diameter **D1**, which is generally the same diameter as the remaining length of the catheter tube **30**. The initial diameter **D1** of the expandable portion **32** may, however, be any desired dimension.

(40) When the catheter device **10** is in the first operating mode, the distal end of the catheter tube **30** can be percutaneously inserted into a blood vessel **50**, as shown in FIGS. **3** and **4**. The illustrated catheter tube **30** is then advanced through the blood vessel **50** along a guide wire **52**, which extends through the catheter device **10**. For example, the guide wire **52** may fully extend through the inner

sleeve **40**, into the internal chamber **22** of the handle body **21**, and exit a rear end of the handle assembly **20** (see FIG. 2). The catheter tube **30** is advanced along the guide wire **52** until the expandable portion **32** is positioned in a narrowed region of the blood vessel **50** caused by atherosclerotic material **54**. Alternatively, the catheter tube **30** can be inserted into the blood vessel **50** and guided therethrough by a delivery catheter (not shown) or any other suitable procedure. During insertion and advancement of the catheter tube **30** through the blood vessel **50**, the optional protective sheath **42** is preferably positioned over the expandable portion **32**, thereby preventing the incising elements **36** from coming into contact with inner walls of the blood vessel **50**.

(41) Once the expandable portion **32** is positioned in the narrowed region of the blood vessel **50**, the incising elements **36** can be exposed by sliding the protective sheath **42** back from the distal end of the catheter tube **30**, as indicated by the direction arrows in FIG. 3. The illustrated protective sheath **42** can be moved in this manner by pulling the flange **44** towards the handle assembly **20**, which is indicated by the direction arrows in FIG. 2.

(42) Referring now to FIGS. 5 through 7, the catheter device **10** is illustrated in a second operating mode. To achieve the second operating mode, the control member **25** is moved from the forward position to the rearward position, as indicated by the direction arrow in FIG. 5. As the control member **25** is moved to the rearward position, the inner sleeve **40** is drawn within the catheter tube **30** thereby reducing the relative length of the inner sleeve **40** with respect to the catheter tube **30**. The distal end of the inner sleeve **40** is attached to the tip member **38**, as described above, causing the expandable portion **32** to become axially compressed between the tip member **38** and the distal end of the catheter tube **30**. As a result, the struts **34A** and **34B** bow or expand outwardly in a generally arcuate fashion thereby defining an opened position. In the opened position, the expandable portion **32** defines a second diameter $D2$ that is larger than the initial diameter $D1$ when the expandable portion **32** is in the closed position. As shown in FIG. 6, the incising elements **36** are respectively positioned along the radially outer most surfaces of the struts **34A** and **34B**. Further, the outer most surfaces of the struts **34A** and **34B** may define a generally flat portion along a length thereof in the opened position, the purpose of which will be explained below, although such is not required. It should be appreciated that the struts **34A** and **34B** can have any lengths such that the expandable portion **32** can achieve a desired overall second diameter $D2$ in the opened position.

(43) During operation of the catheter device **10**, the second diameter $D2$ can be increased or decreased by selective movement of the control member **25** between the forward and rearward positions. For example, a larger second diameter $D2$ can be achieved by moving the control member **25** further towards the rearward position. Conversely, a smaller second diameter $D2$ can be achieved by moving the control member **25** further towards the forward position. The visual indicator **24A** can be used to identify the instantaneous second diameter $D2$ of the expandable portion **32**. Alternatively (or in addition), the struts **34A** and **34B** may be biased in the opened position so as to automatically expand outwardly to the second diameter $D2$ when the protective sheath **42** is slid back from the expandable portion **32**. As such, sliding movement of the protective sheath **42** relative to the struts **34A** and **34B** can be used to selectively control the second diameter $D2$. In this configuration, the inner sleeve **40** and the movable components of the handle assembly **20** may not be necessary.

(44) When the catheter device **10** is in the second operating mode, the expandable portion **32** can be pulled along the guide wire **52** through the narrowed region of the blood vessel **50**. This can be accomplished by pulling on the handle assembly **20**. In doing so, the incising elements **36** engage the atherosclerotic material **54** and create longitudinal incisions **56** therein. As shown in FIGS. 6 and 7, the outer surface area of the arcuate shaped struts **34A** and **34B**, which is adjacent to the incising element **36**, is configured to ride along a surface of the atherosclerotic material **54**, thereby limiting the depth of the incisions **56** and preventing the incising members **36** from cutting the walls of the blood vessel **50**. The expandable portion **32** can be moved any distance along the guide

wire **52** to create incisions **56** having any desired length. After the incisions **56** are made in the atherosclerotic material **54**, the catheter device **10** can be returned to the first operating mode (shown in FIGS. **1** through **4**) by moving the control member **25** to the forward position. In doing so, the expandable portion **32** returns to the closed position. The protective sheath **42** can be slid over the expandable portion **32** and the catheter tube **30** may be removed from the blood vessel **50**. (45) Alternatively, the catheter device **10** can be used to create additional incisions **56** in the atherosclerotic material **54**. For example, after the catheter device **10** has been returned to the first operating mode, the expandable portion **32** can be relocated within the narrowed region of the blood vessel **50**. The catheter tube **30** can then be rotated within the blood vessel **50** by rotating the handle assembly **20** so as to align the incising elements **36** with other portions of the atherosclerotic material **54**. The previous steps can then be repeated any number of times to make multiple passes through the narrowed region of the blood vessel **50** and create additional incisions in the atherosclerotic material **54**.

(46) Thus, it should be appreciated that the illustrated catheter device **10** is advantageous in many respects. In one example, the second diameter **D2** of the expandable portion **32** can be selectively controlled by operation of the handle assembly **20** or by sliding movement of the protective sheath **42**. This enables the catheter device **10** to be adapted for use in blood vessels **50** of different sizes or varying diameters. In another example, the illustrated catheter device **10** can apply varying magnitudes of radial forces to the atherosclerotic material **54** by controlling the amount of force being applied to the control member **25** on the handle assembly **20**. This enables the catheter device **10** to generate sufficient radial force to create incisions **56** in atherosclerotic material **54** while reducing the potential for tearing the walls of the blood vessel **50**. In yet another example, the catheter device **10** can be used to make any number of passes during a single procedure to make multiple incisions **56** in atherosclerotic material **54** of varying lengths and shapes.

(47) Referring now to FIGS. **8** through **10**, there is illustrated a catheter tube **130** having an expandable portion **132**, in accordance with a second embodiment of this invention. The catheter tube **130** and the expandable portion **132** may include any structural features as described and illustrated above in the previous embodiment, although such is not required. Similar features have been numbered with common reference numerals but have been increased by 100 (i.e., **110**, **120**, **130**, etc.). It should be appreciated that similar features are structured similarly, operate similarly, and/or have the same function unless otherwise indicated by the drawings or this specification.

(48) For example, the catheter tube **130** may extend from a handle assembly (not shown) as described above in the first embodiment. The expandable portion **132** is provided on a distal end of the catheter tube **130** and may include a tip member **138**. The catheter tube **130** may also include an inner sleeve **140** and a protective sheath (not shown), which is also described above in the first embodiment.

(49) In the illustrated embodiment, however, the expandable portion **132** includes four struts **134A**, **134B**, **134C**, and **134D** that are respectively separated by four longitudinally extending slits **135A**, **135B**, **135C**, and **135D**. The illustrated struts **134A**, **134B**, **134C**, and **134D** each include an incising element **136**, although such is not required. It should be appreciated that the expandable portion **132** may have any number or configuration of struts and incising elements as desired.

(50) As shown in FIG. **8**, the illustrated expandable portion **132** further includes recessed portions **160** that respectively extend into the outer surfaces of the struts **134A**, **134B**, **134C**, and **134D**. For example, the struts **134A**, **134B**, **134C**, and **134D** can be slightly bowed inwardly toward the inner sleeve **140** when in the closed position or, alternatively, may have a reduced thickness along a central portion thereof to create the recessed portions **160**. The illustrated incising elements **136** are respectively disposed within the recessed portions **160**. Thus, when the catheter tube **130** is inserted into a blood vessel, as described above, the recessed portions **160** help to prevent the incising elements **136** from coming into contact with inner walls of the blood vessel. On the other hand, when the expandable portion **132** is expanded to an opened position, as explained below, the

incising elements **136** become exposed from the recessed portions **160**. It should be appreciated that the recessed portions **160** can eliminate or reduce the need for the protective sheath (not shown). The guide wire **152** may extend through the entire device.

(51) The expandable portion **132** can be operated between a closed position (shown in FIG. **8**) and an opened position (shown in FIGS. **9** and **10**) by selective movement of the inner sleeve **140** relative to the catheter tube **130**, as described above in the first embodiment. Alternatively (or in addition), the struts **134A**, **134B**, **134C**, and **134D** can be biased in the opened position. In such an embodiment, the protective sheath (not shown) can be used to effect movement of the expandable portion **132** between the closed position and the opened position.

(52) Referring now to FIGS. **11** through **13**, there is illustrated a catheter tube **230** having an expandable portion **232**, in accordance with a third embodiment of this invention. The catheter tube **230** and the expandable portion **232** may include any structural features as described and illustrated above in the previous embodiments, although such is not required. Similar features have been numbered with common reference numerals but have been increased by 200 (i.e., 210, 220, 230, etc.). It should be appreciated that similar features are structured similarly, operate similarly, and/or have the same function unless otherwise indicated by the drawings or this specification.

(53) For example, the catheter tube **230** may extend from a handle assembly (not shown) as described above in the first embodiment. The expandable portion **232** is provided on a distal end of the catheter tube **230** and includes a pair of struts **234A** and **234B** that are separated by a pair of longitudinally extending slits **235A** and **235B**. The catheter tube **230** may also include a tip member **238**, an inner sleeve **240**, and a protective sheath (not shown), which is described above in the first embodiment. The guide wire **252** may extend through the entire device.

(54) In the illustrated embodiment, however, the expandable portion **232** includes a first pair of weakened regions **237A**, **237B** and a second pair of weakened regions **239A**, **239B** that are respectively located at opposite ends of the struts **234A** and **234B**. The illustrated weakened regions **237A**, **237B** and **239A**, **239B** are formed by enlarged apertures that extend through side walls of the expandable portion **232** that function as hinges. The weakened regions **237A**, **237B** and **239A**, **239B** may help reduce the amount of bending stress in the side walls of the expandable portion **232** when the struts **234A** and **234B** are moved to an opened position. The struts **234A** and **234B** may include any number or configuration of weakened regions. Further, it should be appreciated that any of the other embodiments in this disclosure may also include weakened regions **237A**, **237B** and **239A**, **239B**.

(55) The illustrated struts **234A** and **234B** remain generally flat along respective lengths thereof in both a closed position (shown in FIG. **11**) and an opened position (shown in FIGS. **12** and **13**) so as to form an apex, although such a configuration is not required. The incising elements **236** are provided along the generally flat portion of the respective struts **234A** and **234B**. As such, the incising elements **236** may also function as stiffening members for increasing the strength of the struts **234A** and **234B**. Further, this configuration can reduce the amount of stress in the connection between the incising elements **236** and the struts **234A** and **234B**, which may otherwise be caused by bowing of the struts **234A** and **234B**.

(56) As shown in FIG. **12**, end portions of the incising elements **236** may extend beyond the apex that is formed by each of the respective struts **234A** and **234B**. This configuration can increase the effective height of the incising elements **236** when the expandable portion **232** is in the opened position. As such, the incising elements **236** may have a reduced height when the expandable portion **232** is in the closed position, which may eliminate the need for the protective sheath (not shown).

(57) The expandable portion **232** can be operated between the closed position and the opened position by selective movement of the inner sleeve **240** relative to the catheter tube **230**, as described above in the first embodiment. Alternatively (or in addition), the struts **234A** and **234B** can be biased in the opened position. In such an embodiment, the protective sheath (not shown) can

be used to effect movement of the expandable portion **232** between the closed position and the opened position.

(58) Referring now to FIGS. **14** through **16**, there is illustrated a catheter tube **330** having an expandable portion **332**, in accordance with a fourth embodiment of this invention. The catheter tube **330** and the expandable portion **332** may include any structural features as described and illustrated above in the previous embodiments, although such is not required. Similar features have been numbered with common reference numerals but have been increased by 300 (i.e., 310, 320, 330, etc.). It should be appreciated that similar features are structured similarly, operate similarly, and/or have the same function unless otherwise indicated by the drawings or this specification.

(59) For example, the catheter tube **330** may extend from a handle assembly (not shown) as described above in the first embodiment. The expandable portion **332** is provided on a distal end of the catheter tube **330** and may include a tip member **338**. The catheter tube **330** may also include an inner sleeve **340** that is attached to the tip member **338** and a protective sheath (not shown), which is also described above in the first embodiment. The guide wire **352** may extend through the entire device.

(60) In the illustrated embodiment, however, the expandable portion **332** includes a pair of struts **334A** and **334B** that are supported thereon in a cantilevered manner (i.e., not attached to one another or to the tip member **338** at their distal ends), the purpose of which will be explained below. The struts **334A** and **334B** are separated by a pair of longitudinally extending slits **335A** and **335B** that extend from the end of the expandable portion **332**. A pair of incising elements **336** is respectively provided along outer surfaces of the struts **334A** and **334B**. It should be appreciated, however, that the expandable portion **332** may have any number or configuration of struts and incising elements as desired.

(61) As shown in FIGS. **15** and **16**, the illustrated struts **334A** and **334B** are supported on the expandable portion **332** so that they can be splayed open in a Y-shaped configuration. For example, the struts **334A** and **334B** can be splayed open by drawing the inner sleeve **340** within the catheter tube **330**, as described above in the first embodiment. In doing so, the tip member **338** slides along the inner surfaces of the struts **334A** and **334B** and pivots them outwardly. Alternatively (or in addition), the struts **334A** and **334B** can be biased in the splayed open position. In such an embodiment, the protective sheath (not shown) can be used to effect movement of the expandable portion **332** between a closed position and the splayed open position.

(62) The struts **334A** and **334B** remain generally flat along their respective lengths in both a closed position (shown in FIG. **14**) and the splayed open position, although such is not required. As such, the incising elements **336** may also function as stiffening members for increasing the strength of the struts **334A** and **334B**. Further, this configuration can reduce the amount of stress in the connection between the incising elements **336** and the struts **334A** and **334B**, which may otherwise be caused by bowing of the struts **334A** and **334B**.

(63) As shown in FIG. **15**, end portions of the incising elements **336** may extend beyond the distal ends of the respective struts **334A** and **334B**. This configuration can increase the effective height of the incising elements **336** when the expandable portion **332** is in the splayed open position. As such, the incising elements **336** may have a reduced height when the expandable portion **332** is in the closed position, which may eliminate the need for the protective sheath (not shown).

(64) FIGS. **17** and **18** illustrate another exemplary embodiment of the present invention. Specifically, FIG. **17** illustrates the device of FIG. **8** with an exemplary embolic protection device **90** installed thereon, while FIG. **18** illustrates the device of FIG. **17** with the expandable portion **132** in the opened position. The embolic protection device **90** may comprise a basket configured to trap emboli for subsequent removal from the treatment site.

(65) The basket may surround the outer surface of the struts **134A**, **1348**, and **134C** as well as cover the gaps between said struts **134A**, **1348**, and **134C** when the expandable portion **132** is in both the closed and the opened positions. To accomplish this, the basket may be configured to stretch and

deform or may comprise excess and/or overlapping material when the expandable portion **132** is in the closed position that unfurls when the expandable portion **132** is in the opened position.

(66) The basket may also cover the front and/or rear of the expandable portion **132**. However, in exemplary embodiments of the present invention, the basket may be open on the proximal end and attached, sealed, bonded, or otherwise adhered to the tip member **138** on the distal end. In this way, the basket creates an opening on the proximal end, and is substantially or partially sealed on the distal end by the combination of the basket, guide wire **152**, inner sleeve **140**, and struts **134A**, **1348**, and **134C**. As will be explained in greater detail herein, the basket may be comprised of a woven material or otherwise comprise a plurality of apertures along the side walls thereof. In exemplary embodiments of the present invention, these apertures may be configured to permit blood to flow therethrough while preventing emboli and other particulate over a specified size from traveling therebeyond. In this way, the blood flow, and any emboli or other particulate contained therein, are forced to enter the basket's opening on the proximal end and leave only if the matter fits through the apertures provided in the sidewalls of the basket.

(67) In exemplary embodiments of the present invention, the basket is comprised of a mesh. The mesh may be of any size, shape, or configuration. The mesh may be comprised of nitinol, polytetrafluoroethylene (PTFE), a metallic, a polymer, or the like and may extend over any section or the entirety of the expandable portion **132**. The mesh may be woven such that the apertures are sized to permit blood (including, for example without limitation, healthy cells, plasma, and platelets) flow therethrough, while trapping emboli and other particulate larger than the apertures provided in the mesh.

(68) The embolic protection device **90** may extend over some or all of the expandable portion **132**. In exemplary embodiments of the present invention, the embolic protection device **90** extends over substantially the distal half of the expandable portion **132** while the incising elements **136** may extend over substantially the proximal half of the expandable portion **132**. Regardless, an expandable hoop **91** may be located on the outer surface of the expandable portion **132** and may assist in securing and preserving the shape of the basket. The expandable hoop **91** may be located at substantially the midline of the expandable portion **132**, though any location is contemplated. The expandable hoop **91** may be comprised of nitinol, PTFE, a metallic, a polymer, or the like and may be configured to expand and collapse when the expandable portion is moved between the opened and the closed positions.

(69) The expandable hoop **91** may be configured to match the outer diameter of the expandable portion **132** when the expandable portion **132** is in both the opened and the closed positions and may be configured to fit inside the sheath **42** if one is being used. In exemplary embodiments of the present invention, the expandable hoop **91** may operate by a telescoping mechanism such that portions of the expandable hoop **91** slide atop one another. In other exemplary embodiments, the expandable hoop **91** may be configured to deform by elongating and reorienting at an increased lateral angle such that the expandable hoop **91** may be placed substantially flush with the outer surface of the expandable portion **132**, when the expandable portion **132** is in the collapsed position and/or is forced inside the sheath **42**.

(70) The expandable hoop **91** may provide an attachment point for the basket. For example, but not to serve as a limitation, in embodiments where the basket is the mesh or other woven material, the mesh may be woven around the expandable hoop **91**. In other exemplary embodiments, the basket may be welded, soldered, adhered, or otherwise bonded to the expandable hoop **91** and/or directly to the struts **134A**, **1348**, and **134C** or other parts of the intravascular catheter device **10**.

(71) In exemplary methods utilizing the present invention, the expandable portion **132** may be placed in the closed position and positioned downstream from the treatment area. The expandable portion **132** may be placed in the opened position. The embolic protection device **90** may be automatically deployed when the expandable portion **132** is placed in the opened position as the basket, and the optional expandable hoop **91** if utilized, may expand when the struts **134A**, **1348**,

and **134C** are placed in the opened position and collapse when the struts **134A**, **1348**, and **134C** are placed in the closed position. However, in exemplary embodiments of the present invention, the expandable hoop **91** may be moved between the closed and the opened positions independently of the expandable portion **132**. For example, but without limitation, the expandable hoop **91** may be configured to automatically expand when removed from the sheath **42**. Regardless, the expandable portion **132** may next be retracted along some or all of the treatment area to facilitate fragmentation of the atherosclerotic plaque. The expandable portion **132** and the embolic protection device **90** may then be placed in the closed position, thereby trapping any emboli caught in the basket during the procedure. This process may be repeated multiple times over the same treatment area or over multiple treatment areas.

(72) Similarly, FIG. **19** illustrates the device of FIG. **11** with another exemplary embolic protection device **90** installed thereon, while FIG. **20** illustrates the device of FIG. **19** with the expandable portion **232** in the opened position. The embolic protection device **90** may be similar to the one shown and described with respect to FIGS. **17** and **18**, and may be similarly operated.

(73) FIGS. **21** and **22** illustrate another exemplary embodiment of the embolic protection device **90**. The basket of the embolic protection device **90** may be a film or covering having a number of apertures **92** located therein. Any size, shape, number, and configuration of the apertures **92** are contemplated. In exemplary embodiments of the present invention, the film or covering is comprised of PTFE, though any material is contemplated. Similarly, the film or covering may be attached to the expandable hoop **91**, though such is not required. The embolic protection device **90** may otherwise be similar to the one shown and described with respect to FIGS. **17-20**, and may be similarly operated.

(74) It is notable that while the embolic protection device **90** is illustrated with respect to the devices of FIGS. **8-9** and **11-12**, it is contemplated that the embolic device **90** may be utilized with any of the embodiments of the present invention as shown and/or described herein.

(75) The principle and mode of operation of this invention have been explained and illustrated in its preferred embodiments. However, it must be understood that this invention may be practiced otherwise than as specifically explained and illustrated without departing from its spirit or scope.

Claims

1. An intravascular catheter device comprising: a flexible catheter tube; an expandable portion secured to a distal end of said flexible catheter tube and comprising struts operable between a first position and a second position where the struts are moved outward from the first position, wherein a longitudinal axis of each of said struts extends in a linear fashion in line with a longitudinal axis of the expandable portion, at least when the expandable portion is in the first position; an incising element provided at, and extending outward from, an outer surface of at least one of the struts such that the outer surface of the at least one of the struts extends laterally beyond the incising element; and an embolic protection device covering at least a portion of the outer surface of each of the struts of the expandable portion.
2. The intravascular catheter device of claim 1 wherein: said incising element is provided at a proximal half of said at least one of the struts; and said embolic protection device is provided at a distal half of said expandable portion.
3. The intravascular catheter device of claim 2 wherein: said incising element, including an upper surface thereof, extends longitudinally in a linear fashion along said longitudinal axis of said at least one of the struts.
4. The intravascular catheter device of claim 3 wherein: a longitudinal axis of said incising element extends along a midline of the outer surface of said at least one of the struts.
5. The intravascular catheter device of claim 1 further comprising: an inner sleeve configured for sliding movement within the flexible catheter tube to move the struts between the first position and

the second position.

6. The intravascular catheter device of claim 5 further comprising: a tip member connecting each of the struts to said inner sleeve at the distal end.

7. The intravascular catheter device of claim 6 wherein: each of the struts are connected to said inner sleeve at a distal end by way of the tip member and the flexible catheter tube at a proximal end so as to bow at least a mid-portion outwardly when the expandable portion is placed in the second position.

8. The intravascular catheter device of claim 6 wherein: each of the struts comprise two sections configured for hinging movement relative to one another and said inner sleeve and said flexible catheter tube so as to form a peak at a mid-portion thereof when the expandable portion is placed in the second position.

9. The intravascular catheter device of claim 6 wherein: the embolic protection device comprises a mesh and an expandable hoop; said mesh is attached to said tip member at a first end and said expandable hoop at a second end; and said mesh surrounds an outer surface of each of the struts.

10. The intravascular catheter device of claim 6 wherein: the embolic protection device comprises a covering, apertures in said covering, and an expandable hoop; said covering is attached to said tip member at a first end and said expandable hoop at a second end; and said covering surrounds an outer surface of each of the struts.

11. The intravascular catheter device of claim 5 further comprising: a tip member connected to said inner sleeve, wherein each of the struts are connected to the flexible catheter tube at a proximal end and are free at a distal end, and where said tip member is configured to ride along inner surfaces of each of said struts to force said struts to splay outward in said second position.

12. The intravascular catheter device of claim 1 further comprising: a sheath configured for movement between a first position where the expandable portion is covered and a second position where the expandable portion is exposed.

13. The intravascular catheter device of claim 12 wherein: the struts are biased in the second position.

14. The intravascular catheter device of claim 1 wherein: said incising element is provided at a longitudinal mid-portion of said at least one of the struts; and said embolic protection devices surrounds a distal portion of said expandable portion.

15. The intravascular catheter device of claim 14 wherein: said incising element extends outward in a radial manner from an outer surface of said at least one of the struts; and a longitudinal axis of said incising element extends along a midline of the outer surface of said at least one of the struts.

16. The intravascular catheter device of claim 1 wherein: a footprint of the outer surface of the at least one of the struts is larger than a footprint of said incising element.

17. An intravascular catheter device comprising: a handle assembly; a flexible catheter tube extending from said handle assembly; an inner sleeve configured for sliding movement within the flexible catheter tube; an expandable portion secured to a distal end of said flexible catheter tube and comprising struts connected at a proximal end thereof to said flexible catheter tube and operable between a first position and a second position where the struts are moved outward from the first position, wherein a longitudinal axis of each of said struts extends in a linear fashion along a longitudinal axis of the expandable portion, and wherein each of the struts are spaced apart from one another, at least when in the second position; a tip member connected to the inner sleeve and each of the struts, wherein said expandable portion is moved between said first position and said second position by axial movement of said tip member by way of said inner sleeve; an incising element provided at, and extending outward from, an outer surface of a proximal half of each of the struts, wherein each of said incising elements extend linearly and lengthwise along a midline of the outer surface of an associated one of the struts such that the outer surface of the associated one of the struts extends laterally beyond the incising element; and an embolic protection device surrounding the outer surface of a distal half of the struts and comprising an expandable hoop

located at a mid-portion of said expandable portion and a mesh extending between said expandable hoop and said tip member.
